

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 8: Profit, Loss and Discount

Topic-wise short notes with solved examples

Cost price, selling price, profit and loss as amounts and percentages, overhead expenses, marked price and discount, successive discounts and tax on sales.

1 Basic terms

- **Cost price (C.P.):** the price at which an article is bought. **Selling price (S.P.):** the price at which it is sold.
- **Overheads:** extra expenses such as transport, repairs and labour. They are **added to the cost price**.
- **Profit (gain)** = S.P. – C.P., when S.P. > C.P.
- **Loss** = C.P. – S.P., when C.P. > S.P.
- Profit and loss percentages are always calculated **on the cost price**.

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100, \quad \text{Loss}\% = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Example 1.1 A table bought for ₹1600 is sold for ₹1840. Find the profit per cent.

SOLUTION

$$\text{Profit} = 240. \frac{240}{1600} \times 100$$

15%

Example 1.2 A man buys a second-hand cycle for ₹2400, spends ₹300 on repairs and sells it for ₹2430. Find his loss per cent.

SOLUTION

$$\text{Total C.P.} = 2700. \text{ Loss} = 270. \frac{270}{2700} \times 100$$

10%

2 Finding S.P. and C.P.

To find	Given a profit of $p\%$	Given a loss of $l\%$
S.P.	$\text{C.P.} \times \frac{100 + p}{100}$	$\text{C.P.} \times \frac{100 - l}{100}$
C.P.	$\text{S.P.} \times \frac{100}{100 + p}$	$\text{S.P.} \times \frac{100}{100 - l}$

Example 2.1 A fan costs ₹1500. At what price must it be sold to gain 12%?

SOLUTION

$$1500 \times \frac{112}{100}$$

₹1680

Example 2.2 By selling a watch for ₹1140 a shopkeeper loses 5%. Find the cost price.

SOLUTION

$$1140 \times \frac{100}{95}$$

₹1200

Example 2.3 By selling an article for ₹720 a man loses 10%. At what price should he sell it to gain 5%?

SOLUTION

$$\text{C.P.} = 720 \times \frac{100}{90} = 800. \text{ New S.P.} = 800 \times \frac{105}{100}.$$

₹840

3 Several articles

- When the C.P. of x articles equals the S.P. of y articles, take the C.P. of one article as ₹1.
- When articles are bought at one rate and sold at another, find the C.P. and S.P. of the **same number** of articles (use the LCM).
- Two articles sold at the **same price**, one at a gain of $x\%$ and the other at a loss of $x\%$: there is always a **loss of** $\frac{x^2}{100}\%$ overall.

Example 3.1 The cost price of 12 pens equals the selling price of 10 pens. Find the profit per cent.

SOLUTION

Let the C.P. of one pen be ₹1. S.P. of 10 pens = 12, C.P. of 10 pens = 10.

$$\frac{2}{10} \times 100 = 20\% \text{ profit}$$

Example 3.2 Lemons are bought at 5 for ₹20 and sold at 4 for ₹20. Find the gain per cent.

SOLUTION

C.P. of one = 4; S.P. of one = 5.

25%

Example 3.3 A man sells two radios for ₹1980 each, gaining 10% on one and losing 10% on the other. Find his overall gain or loss.

SOLUTION

$$\text{C.P. of the first} = 1980 \times \frac{100}{110} = 1800; \text{ of the second} = 1980 \times \frac{100}{90} = 2200.$$

$$\text{Total C.P.} = 4000; \text{ total S.P.} = 3960.$$

A loss of ₹40, that is 1%

4 Discount

- **Marked price (M.P.)** or list price: the price printed on the article.
- **Discount:** a reduction offered on the marked price. Discount is always calculated **on the marked price**.
- $\text{S.P.} = \text{M.P.} - \text{discount} = \text{M.P.} \times \frac{100 - d}{100}$, where d is the discount per cent.

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

Example 4.1 A shirt marked ₹800 is sold at a discount of 15%. Find the selling price.

SOLUTION

$$800 \times \frac{85}{100}$$

₹680

Example 4.2 After a discount of 20%, a bag is sold for ₹960. Find its marked price.

SOLUTION

$$960 \times \frac{100}{80}$$

₹1200

Example 4.3 An article marked ₹1250 is sold for ₹1100. Find the discount per cent.

SOLUTION

$$\frac{150}{1250} \times 100$$

12%

5 Marked price, discount and profit together

- First get the S.P. from the M.P. and the discount; then compare the S.P. with the C.P.
- To fix the marked price: find the required S.P. from the C.P. and profit, then divide by $\frac{100 - d}{100}$.

Example 5.1 A shopkeeper buys an article for ₹600 and marks it 30% above the cost price. He allows a discount of 10%. Find his profit per cent.

SOLUTION

$$\text{M.P.} = 780. \text{ S.P.} = 780 \times 0.9 = 702. \text{ Profit} = 102.$$

17%

Example 5.2 How much above the cost price must a trader mark his goods so that after allowing a discount of 20% he still gains 12%?

SOLUTION

$$\text{Let C.P.} = 100. \text{ Required S.P.} = 112. \text{ M.P.} = 112 \times \frac{100}{80} = 140.$$

40% above the cost price

6 Successive discounts

- Two discounts of $d_1\%$ and $d_2\%$ one after the other: $\text{S.P.} = \text{M.P.} \times \frac{100 - d_1}{100} \times \frac{100 - d_2}{100}$.
- They are equivalent to a single discount of $\left(d_1 + d_2 - \frac{d_1 d_2}{100}\right)\%$, which is less than $d_1 + d_2$.

Example 6.1 Find the single discount equivalent to successive discounts of 20% and 10%.

SOLUTION

$$20 + 10 - \frac{200}{100}$$

28%

Example 6.2 A washing machine marked ₹15,000 is sold with successive discounts of 10% and 5%. Find the selling price.

SOLUTION

$$15000 \times 0.9 \times 0.95$$

₹12,825

7 Tax on sales

- A tax such as GST is charged as a percentage of the **selling price** and added to the bill.
- Amount paid = $\text{S.P.} \times \left(1 + \frac{\text{tax}\%}{100}\right)$.
- If the price **includes** tax, the price before tax = $\text{amount paid} \div \left(1 + \frac{\text{tax}\%}{100}\right)$.

Example 7.1 A pair of shoes is priced at ₹1500 and the tax is 12%. Find the amount to be paid.

SOLUTION

$$1500 \times 1.12$$

₹1680

Example 7.2 A fan is bought for ₹2360 including tax at 18%. Find its price before tax.

SOLUTION

$$2360 \div 1.18$$

₹2000

8 Common mistakes

- Calculating profit% or loss% on the selling price.
- Calculating discount on the cost price instead of the marked price.
- Leaving out the overheads from the cost price.
- Adding successive discounts: 20% and 10% make 28%, not 30%.
- Using $100 + l$ for a loss or $100 - p$ for a profit.
- Finding the C.P. by taking the percentage of the S.P.